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Roll No

AU/ME-504 (GS)

B.E. V Semester Examination, June 2020

Grading System (GS)

Machine Component Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Give the standard design procedure of an internal expanding shoe brake, mentioning the assumptions clearly along with the design equations.
2. Explain:
 - a) S-N curve
 - b) Surge in springs
3. What is Nip of leaf spring?
4. With the help of a neat sketch, bring out clearly the difference between Goodman diagram and modified Goodman diagram. Also explain to which types of materials Gerber's parabolic relation and Soderberg's equation is applicable.
5. Discuss the steps to design a simple screw jack and its components.
6. Define the following terms :
 - i) Size factor
 - ii) Surface factor
 - iii) Cumulative fatigue damage factor
7. What is the condition of self-locking in differential band brake? Why should it be avoided in speed-control brakes? What is back-stop band brake?

OR

A propeller shaft is required to transmit 45 kW power at 500 rpm. It is a hollow shaft, having inside diameter 0.6 times of outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm^2 . Calculate the inside and outside diameters of the shafts.

8. Write short notes on any two of the following:
 - i) Design for finite life
 - ii) Centrifugal clutches
 - iii) Band and block brake
 - iv) Equivalent bearing load and load - life relationship
